



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 03

1. Title: Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code), Web Browser (Google Chrome / Microsoft Edge), HTML, JavaScript, Windows Operating System.

3. Theory:

This program demonstrates the use of control structures and form validation in JavaScript. An HTML form is used to accept marks from the user. Validation checks whether the input is empty and ensures the entered marks are between 0 and 100. If the input is invalid, an alert message is displayed. If the input is valid, if-else control statements are used to determine the grade (A, B, C, D, or F) based on the entered marks, and the result is displayed on the webpage.

4. OUTPUT:



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File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/Practical%203%20java%20

Vithika Saoji

Student Grading System

Submit

Marks: 90
Grade: A+

5. Result/Conclusion: The grading system was successfully implemented using HTML, CSS, and JavaScript. The program validates user input, prevents invalid entries, and accurately assigns grades based on the entered marks, demonstrating the effective use of form validation and control structures.



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Case Study No.: 03

1. Case Study Title: Write a JavaScript program that accepts student marks and displays the scholarship category.

2. Case Study Program Code:

```
JS Practical2.js  index.html  Case study 2 java.html  JS  ||  ?  ↓  ↑  ↺  □

P3 case study.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Scholarship Eligibility</title>
7  <style>
8      body{
9          font-family: Arial, sans-serif;
10         background-color: #064c96;
11         display: flex;
12         justify-content: center;
13         align-items: center;
14         height: 100vh;
15     }
16     .container{
17         background: rgb(217, 4, 4);
18         padding: 30px;
19         width: 350px;
20         text-align: center;
21         border-radius: 10px;
22         box-shadow: 0 0 10px gray;
23     }
24     h1{
25         color: navy;
26     }
27     input{
28         width: 90%;
29         padding: 10px;
30         margin: 15px 0;
31         font-size: 16px;
32     }
```

```
33     button{
34         padding: 10px 20px;
35         background: navy;
36         color: white;
37         border: none;
38         border-radius: 5px;
39         cursor: pointer;
40         font-size: 16px;
41     }
42     button:hover{
43         background: blue;
44     }
45     #result{
46         margin-top: 20px;
47         font-size: 20px;
48         font-weight: bold;
49         color: green;
```



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```
51 </style>
52 </head>
53 <body>
54 <div class="container">
55 <h1>Vithika Saoji</h1>
56 <h2>Scholarship Eligibility</h2>
57 <input type="number" id="marks" placeholder="Enter Marks">
58 <br>
59 <button onclick="checkScholarship()">Submit</button>
60 <div id="result"></div>
61 </div>
62 <script>
63 function checkScholarship()
64 {
65     let marks = document.getElementById("marks").value;
66     if(marks=="")
67     {
68         alert("Enter Marks");
69         return;
70     }
71     marks = Number(marks);
72     if(marks<0 || marks>100)
73     {
74         alert("Enter Valid Marks");
75         return;
76     }
77     let scholarship;
78     if(marks<50)
79     {
80         scholarship = "Not Eligible";
81     }
82     else if(marks>=50 && marks<=69)
83     {
84         scholarship = "Bronze Scholarship";
85     }
86     else if(marks>=70 && marks<=84)
87     {
88         scholarship = "Silver Scholarship";
```

```
89     }
90     else if(marks>=85 && marks<=94)
91     {
92         scholarship = "Gold Scholarship";
93     }
94     else
95     {
96         scholarship = "Platinum Scholarship";
97     }
98     document.getElementById("result").innerHTML =
99     "Marks: " + marks + "<br>Scholarship: " + scholarship;
100 }
101 </script>
102 </body>
103 </html>
```

3. Output:

File C:/Users/vithi/OneDrive/Desktop/vithikasaoji%20javascript/P3%20case%20study.html

This page says

Enter Marks

OK

Vithika Saoji

Scholarship Eligibility

